Shadowseed

Manual

Version 0.12.4 · Research Preview · October 2026

PolyForm Noncommercial License 1.0.0 · github.com/E-AI-MODEL/shadowseed-pro

About this manual

Shadowseed is a chat application that keeps track of what is still open in a conversation. This manual explains how to install it, how to work with it and how it works inside. It is written for five kinds of reader. You do not need to read all of it:

You are…	Read
User: you want to work with Shadowseed	Parts 1, 2 and 3
Tester: you try out a new version and report findings	Parts 1, 2, 3 and 4
Researcher: you study Shadow Seed Learning	Parts 1, 3 and 5
Developer: you build on Shadowseed or change it	Parts 1, 5 and 6
Specialist in security, privacy or operations	Parts 1, 6 and 7

Buttons and fields are shown in **bold**, exactly as they appear on screen. A glossary at the end lists the main terms. A Dutch edition is available as `shadowseed-handleiding-nl`.

> **Caution.** Shadowseed is a *Research Preview*: it works and is tested, but it is not meant for decisions with serious consequences, such as in health care, law, finance or employment. Chapter 16 lists the limitations.

Contents

Getting to know Shadowseed 3

1 What Shadowseed does 4

2 A tour 4

Installation 6

3 What you need 7

4 Installing and starting 7

5 Setting up a local model 8

Working with Shadowseed 10

6 Starting a conversation 11

7 Reading seeds 12

8 Confirming a seed with a source 14

9 When a seed takes part 15

10 Contradicting a seed 17

11 Authority modes 18

12 On a phone-sized screen 19

13 Using OpenAI 19

14 Data, backups and privacy 20

15 Troubleshooting 21

16 Limitations 22

Testing 23

17 A test session 24

Research 25

18 The SSL model 26

19 Showing that SSL had an effect 27

20 Studies and evidence 27

Development 28

21 Working from source 29

22 Architecture 30

23 Command line 31

24 Web API 31

25 Builds and releases 32

Security and operations 33

26 Security model 34

27 Running reliably 34

28 Settings 35

Appendices 36

A Glossary 37

B Frequently asked questions 37

C About this document 38

PART 1

Getting to know Shadowseed

What Shadowseed does, and a tour based on an example conversation.

IN THIS PART

1 What Shadowseed does

2 A tour

1 What Shadowseed does

An ordinary chat forgets what was left open along the way. You ask something, the model answers, and an assumption nobody checked drops out of sight. Shadowseed holds on to those open points.

With every answer, Shadowseed looks for questions that are still unanswered: a missing figure, an assumption that was not checked, a situation that was not covered. It stores such a point as a **seed**, a short note in the margin of the conversation. Together these notes form the **Shadow memory**.

A seed is a hunch, not a fact. That is why a new seed has no influence on the answers at all. A seed gains influence only when you have checked a source that supports it and the **Validation Gate** has accepted that source. Even then, Shadowseed uses the seed only when it fits the question you are asking at that moment.

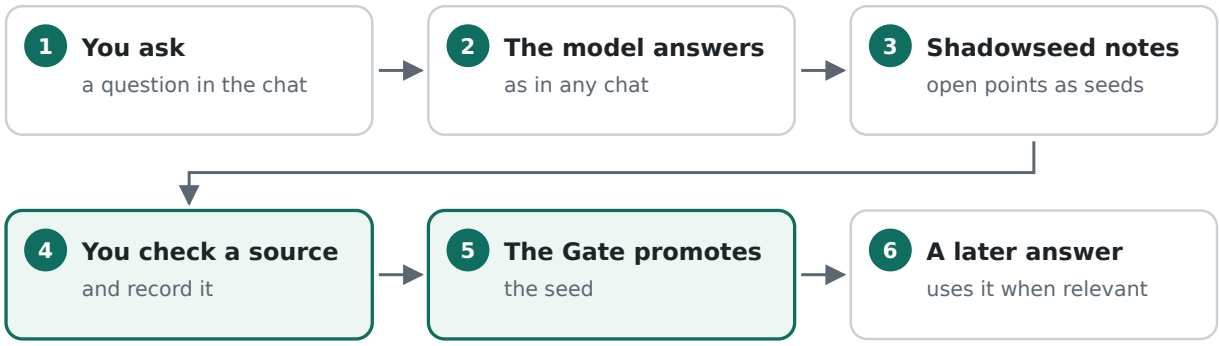


Figure 1 — From question to influence. Only the green steps give a seed influence, and you are in charge of those.

This way of working is called **Shadow Seed Learning (SSL)**. Three principles shape the design:

- **Notice widely, admit sparingly.** Shadowseed notes many open points but lets few of them steer.
- **Influence comes from evidence, not from repetition.** A point that keeps coming back is interesting, but that does not make it true.
- **Everything can be traced.** Every seed has a timeline with every observation and every decision.

Shadowseed does not retrain the language model. The model stays the same; only the memory around it grows.

What Shadowseed does not promise. A seed can be wrong. A source you mark as checked can itself be wrong. And whether SSL makes answers better is still being studied (chapter 20).

2 A tour

This manual follows one example conversation throughout. A school wants a backup plan for its student administration and asks Shadowseed for advice. After three questions the screen looks like this:

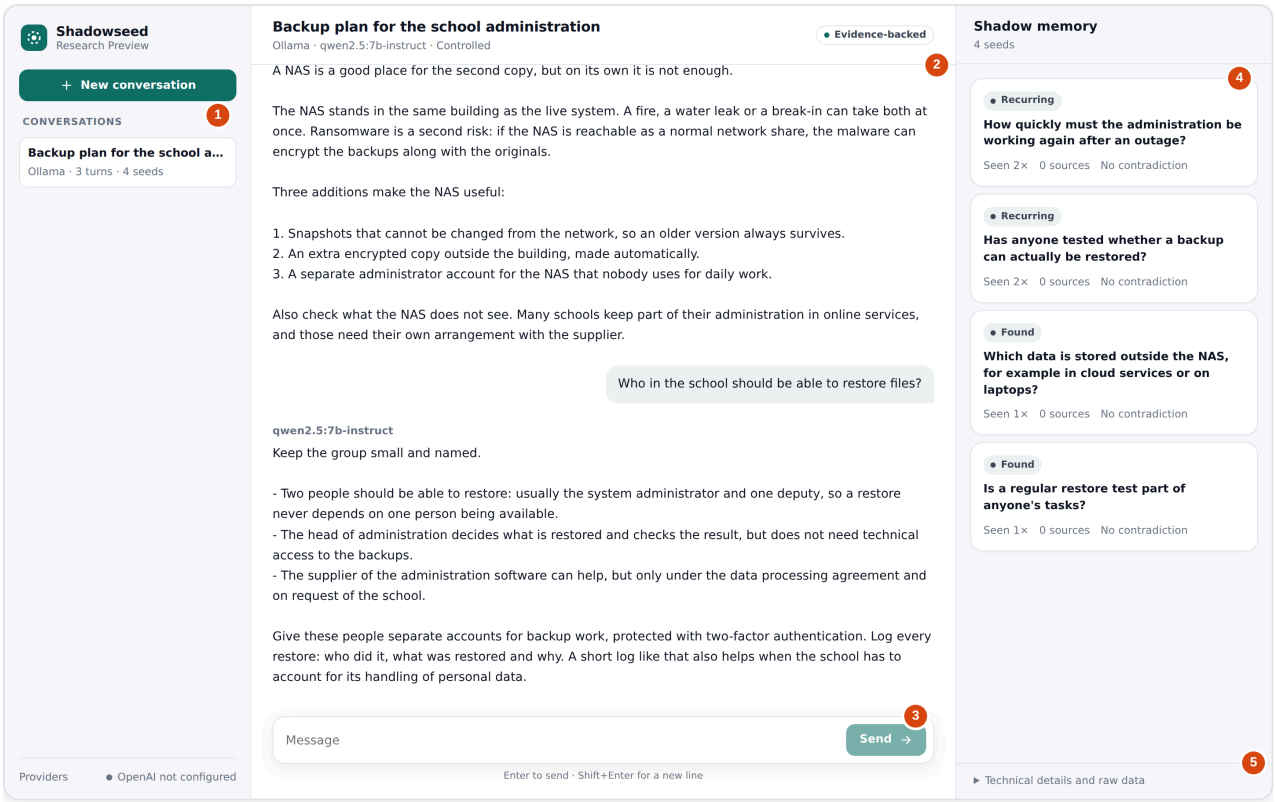


Figure 2 — The example conversation after three questions. The numbers refer to the explanation below.

1. **Conversations.** Your conversations, with the model, the number of turns and the number of seeds. At the top, **New conversation** starts a new one.
2. **The chat.** Here you talk to the model, as in any chat application. The header shows the title, the model and the authority mode (chapter 11).
3. **The message box.** Type your question and press **Enter**. **Shift+Enter** starts a new line.
4. **The Shadow memory.** The seeds of this conversation. Each card shows the status, the text and how often the seed was seen.
5. **Technical details.** For anyone who wants to look under the hood (chapter 9).

In the example Shadowseed noted four open points, and two of them came back in a later turn. *How quickly must the administration be working again after an outage?* is one of them: the advice depends on it, but nobody mentioned it.

Part 3 continues the story. The school records a board decision as a source (chapter 8), and the next answer takes that decision into account (chapter 9).

> **About the examples.** The screenshots come from Shadowseed 0.12.4. Seeds, statuses, Gate decisions and the timeline were produced by the program itself. The model's answers were written in advance for this manual, so the examples are readable and repeatable. A real model words things differently and does not find this many seeds in every conversation.

PART 2

Installation

Shadowseed runs on your own computer. You need no account, and your conversations stay local.

IN THIS PART

3 What you need

4 Installing and starting

5 Setting up a local model

3 What you need

	Minimum	Recommended
Operating system	Windows 10 or 11 (64-bit), macOS on Apple silicon (M1 or later), Linux x86_64	
Memory	8 GB	16 GB or more
Disk space	200 MB for Shadowseed, plus about 5 GB per model	
Graphics card	Not required	A GPU makes answers much faster
Ollama	Version 0.11.10	The latest version
Browser	A recent Chrome, Edge, Firefox or Safari	

You need the internet only to download Shadowseed and the models.

How fast is it? That depends on your computer and the model. Without a suitable graphics card, a medium-sized model (7B) takes two to three minutes per answer. A GPU or a smaller model is much faster. The offline demo needs no model and responds at once.

4 Installing and starting

Download

On the project's GitHub release page, download the file for your system:

System	File
Windows	`shadowseed-web-0.12.4-windows-amd64.zip`
macOS (Apple silicon)	`shadowseed-web-0.12.4-darwin-arm64.zip`
Linux	`shadowseed-web-0.12.4-linux-x86_64.tar.gz`

Verify the download

To be sure your file is exactly the published file, also download `SHA256SUMS` and compute the checksum:

```
Windows (PowerShell)   Get-FileHash .\shadowseed-web-0.12.4-windows-amd64.zip
macOS                  shasum -a 256 shadowseed-web-0.12.4-darwin-arm64.zip
Linux                  sha256sum shadowseed-web-0.12.4-linux-x86_64.tar.gz
```

The result must match the line for your file in `SHA256SUMS` character for character.

Start on Windows

1. Right-click the ZIP file and choose *Extract All*.
2. Open the extracted folder, then the `Shadowseed` folder inside it.
3. Double-click `Shadowseed.exe`.

Windows may warn that the app is unrecognised, because it is not signed with a Microsoft certificate. Choose *More info*, then *Run anyway*.

Start on macOS

1. Double-click the ZIP file to extract it.
2. Right-click `Shadowseed`, choose *Open* and confirm.

macOS blocks the first start because the app is not notarised by Apple. If opening with a right-click does not work, run this once in Terminal:

```
xattr -dr com.apple.quarantine /path/to/Shadowseed.app
```

Start on Linux

```
tar -xzf shadowseed-web-0.12.4-linux-x86_64.tar.gz
./Shadowseed/Shadowseed
```

After starting

Shadowseed opens your browser at an address such as `http://127.0.0.1:8765/`. That address exists only on your own computer; nobody else can reach it. If port 8765 is already in use, Shadowseed picks another one.

Keep the Shadowseed window open while you work. If you close it, the program stops. Your conversations are kept.

> **Tip.** If the browser does not open by itself, copy the address from the Shadowseed window into your browser.

---

5 Setting up a local model

For real conversations Shadowseed needs a language model. It uses **Ollama** for that, a free program that runs models on your own computer.

1. Install Ollama from ollama.com and start it.
2. Open a terminal and download a chat model, for example: `ollama pull qwen2.5:7b-instruct`
3. Download the model Shadowseed uses to compare texts: `ollama pull embeddinggemma`

4. Check that both models are there: `ollama list`

Which chat model? Any chat model Ollama can run works. Larger models give better answers and find more useful seeds, but need more memory and time. From our measurements:

Model	Seeds	Useful
Qwen 2.5 7B on an ordinary computer	about 1 per 8 turns	about a quarter
Large hosted models (GPT-5, GPT-6.1-sol)	much more often	51 of 60 in a sample

If you mainly want to see how Shadowseed works, a small model or the offline demo is enough. If you want useful seeds, choose the largest model your computer can handle.

PART 3

Working with Shadowseed

From the first conversation to managing your data.

IN THIS PART

6 Starting a conversation

7 Reading seeds

8 Confirming a seed with a source

9 When a seed takes part

10 Contradicting a seed

11 Authority modes

12 On a phone-sized screen

13 Using OpenAI

14 Data, backups and privacy

15 Troubleshooting

16 Limitations

6 Starting a conversation

Click **New conversation** at the top left. A form appears:

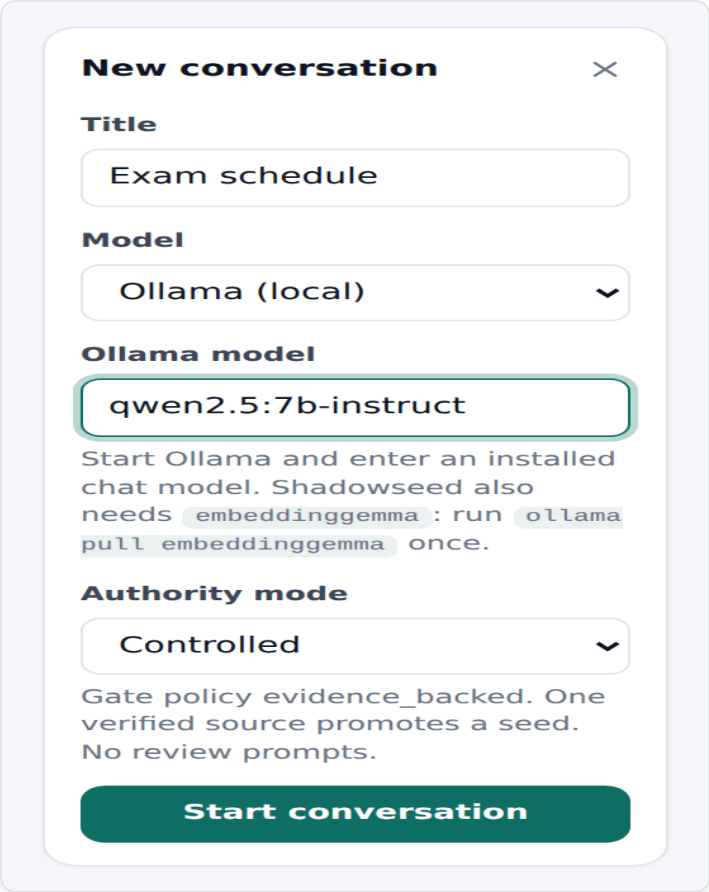


Figure 3 — A new conversation with a local model.

1. Give the conversation a **Title**.
2. Choose a **Model**: - *Offline demo (no model)* to explore the controls. The answers are simple echoes and the seeds are marked `[FIXTURE]`. - *Ollama (local)* for real conversations.
3. Under **Ollama model**, enter the name exactly as `ollama list` shows it, for example `qwen2.5:7b-instruct`.
4. Leave **Authority mode** on *Controlled*. Chapter 11 explains the other modes.
5. Click **Start conversation**.

If Ollama is not running, or `embeddinggemma` is missing, Shadowseed says what is wrong and does not create the conversation. Fix it and try again.

Then ask your question as you would in any chat. Seeds appear in the Shadow memory by themselves. The longer the conversation, the more likely an open point comes back.

7 Reading seeds

After three questions, the example has four seeds in the Shadow memory:

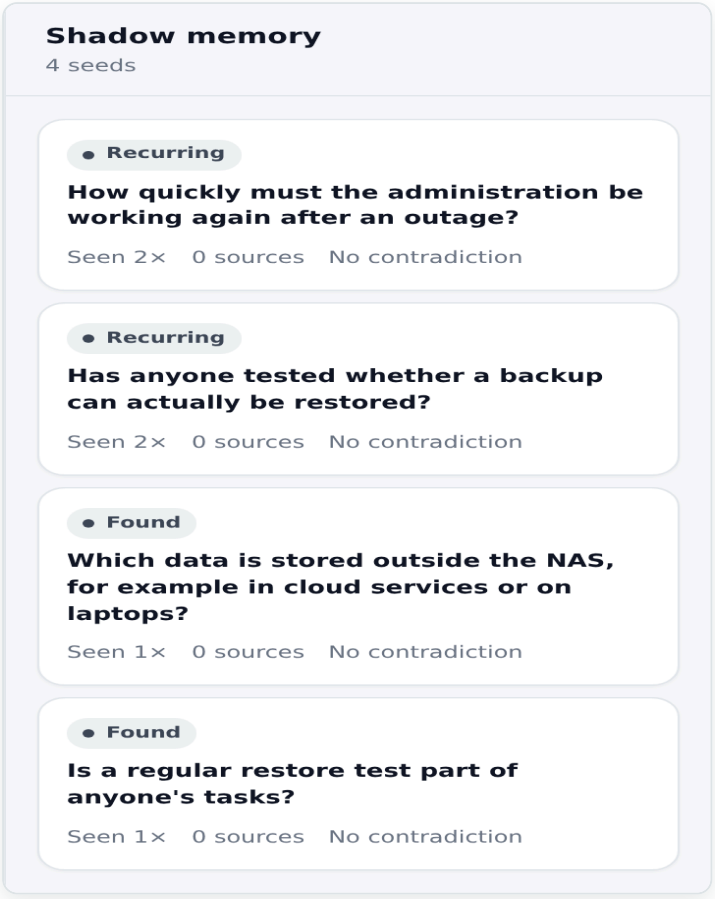


Figure 4 — The Shadow memory. Each card shows the status, the text, how often the seed was seen, the number of verified sources and whether it is contradicted.

Each seed is phrased as an open question about the conversation. Its **status** tells you what the seed may do right now:

Status	Meaning	Influence on answers
Found	Noticed once.	No
Recurring	Noticed again in a later turn.	No
Supported	Has a verified source but has not passed the Gate.	No
Validated	Has weight from the Gate but is not promoted.	No
Promoted	Passed the Gate. May take part when it fits the question.	Yes, when relevant
Used	Promoted and actually used at least once.	Yes, when relevant
Blocked	A contradiction is open.	No
Dormant	Not seen for a while. Becomes active again when the topic returns.	No

Status	Meaning	Influence on answers
Expired	Faded out. Kept for the record.	No

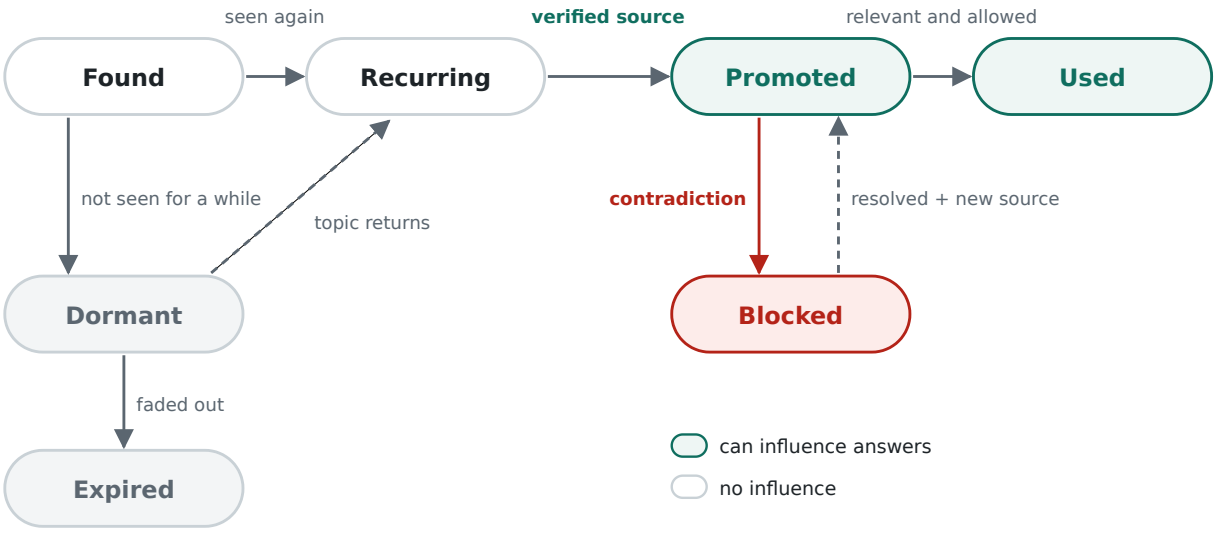


Figure 5 — How a seed changes status. With one verified source a seed can also go straight from Found to Promoted, and any seed can be contradicted.

How Shadowseed recognises recurrence

Two seeds can share a topic without asking the same question. So Shadowseed asks the model itself whether a new note is the same open question as an earlier one. Only "the same question" counts as recurrence. In the example the model recognised *How long may the school be without the administration if the NAS fails?* as the same question as *How quickly must the administration be working again after an outage?*

Recurrence is an observation, not evidence. A seed that comes back ten times still has no influence in the default mode.

Opening a seed

Click a card to open the seed. Six counters appear at the top:

Counter	Meaning
Seen	In how many turns the seed was observed
Verified sources	How many different checked sources support it
Contradiction	Whether a contradiction is open
Weight	How much weight the Gate gave it, from 0 to 1. New seeds start at 0
Used in answers	How often it influenced an answer
Trace	How recently the seed was present; this number falls when it does not return

Below them are the forms for evidence and contradiction, the timeline and the raw data.

8 Confirming a seed with a source

The school in the example knows the answer to the open question: the board has decided that the administration must be working again within one working day. This is how you record that:

1. Open the seed.
2. Under **Add verified evidence**, enter a stable reference in **Source reference**: a URL, a DOI or, as here, a document that can be found.
3. Under **Note**, explain in one sentence why the source supports the seed.
4. Tick **I checked this source and confirm it supports this seed.**
5. Click **Record evidence**.



Figure 6 — Recording a source for a recurring seed.

The Gate assesses the source at once. In the default mode one verified source is enough: the seed becomes **Promoted** with weight 0.20.

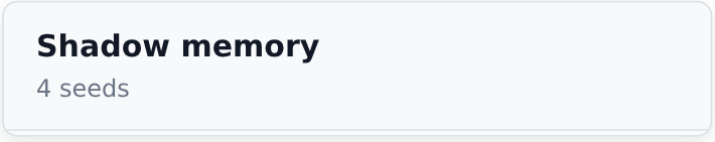


Figure 7 — The same seed after one verified source.

> **Caution.** By ticking the box you declare that you read the source yourself. Shadowseed records that declaration in the audit ledger. Shadowseed does not check whether the source is right; that remains your responsibility.

Four rules keep the Gate reliable:

- **Only tick what you have read.**
- **One source counts once.** Entering the same reference again adds no weight.
- **A model answer is not a source.** Refer to the original document, not to what the model said about it.
- **Recurrence is not a source.** A seed that keeps coming back still needs a source.

9 When a seed takes part

A promoted seed takes part only when it fits the question you ask. In the example the school next asks how often the backup should run. The seed about recovery time fits that question, and the answer takes it into account:

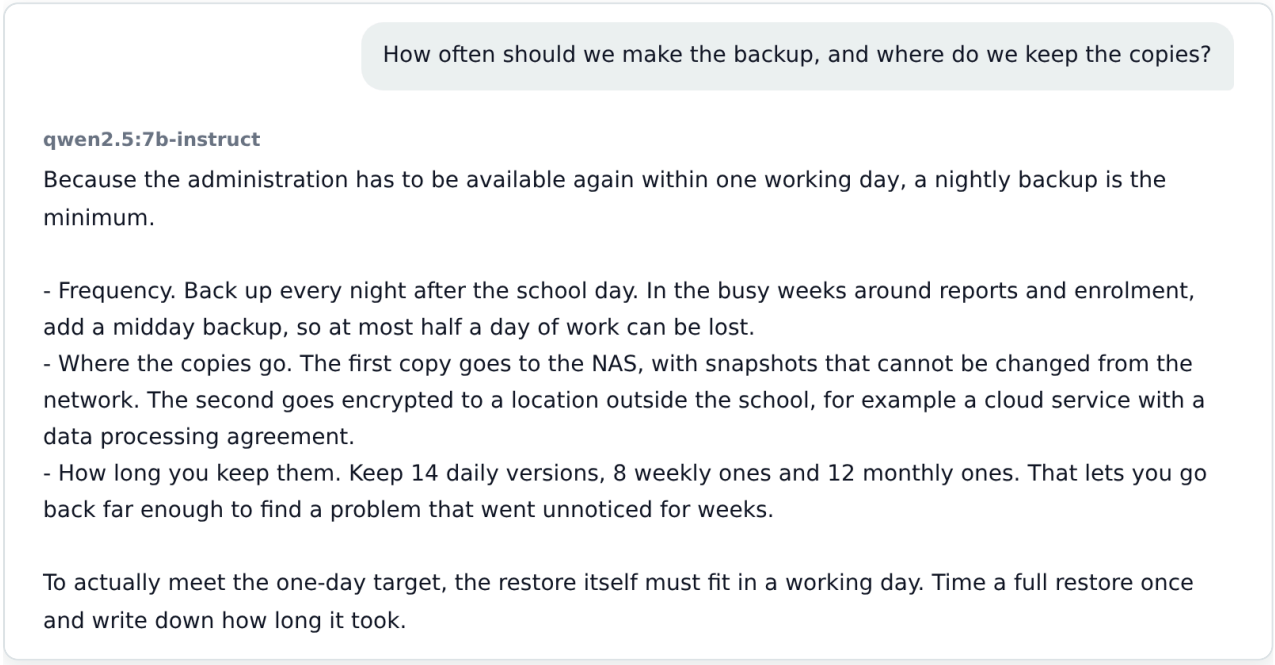


Figure 8 — The answer uses the recorded decision: recovery within one working day.

The seed's status becomes **Used**, and the card shows how often it was used:

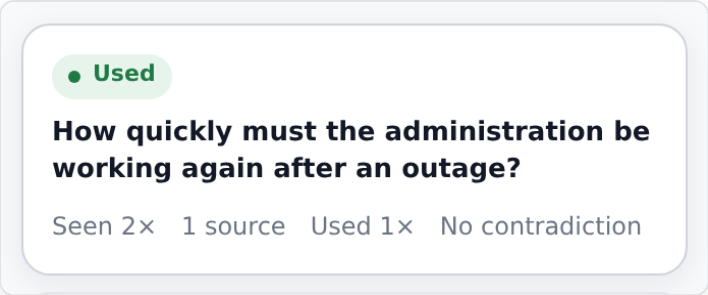


Figure 9 — After use the seed shows as Used.

Shadowseed lets a seed take part only when, at the moment of your question, all of the following hold:

1. the seed is promoted and has weight;
2. that promotion rests on a valid Gate decision;
3. no contradiction is open;
4. the seed fits your current question;
5. the last check just before the answer (*point of use*) allows it.

So *Promoted* does not mean *always used*. If you ask the example conversation about passwords, the seed about recovery time stays silent. That is intended.

The seed reaches the model as background, explicitly marked as unverified context. The model is not told that it is a fact. And when a seed has been used in an answer, new seeds from that same answer are not stored. That way Shadowseed does not learn from its own influence.

The timeline

At the bottom of the seed is the **Timeline**. It lists every observation, validation, Gate decision and use, newest first. Expand a line for the full record.

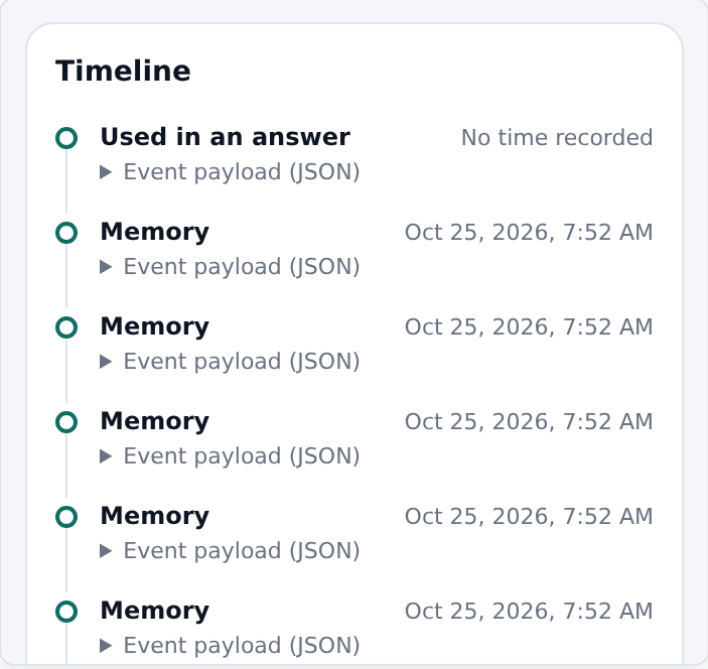


Figure 10 — The timeline of the used seed.

Technical details

Below the Shadow memory, **Technical details and raw data** opens the conversation's data: the Gate policy, the authority profile, a fingerprint of all settings (*behavior digest*) and the complete raw data of the conversation and the last turn. **Copy** copies it, for example for a problem report.

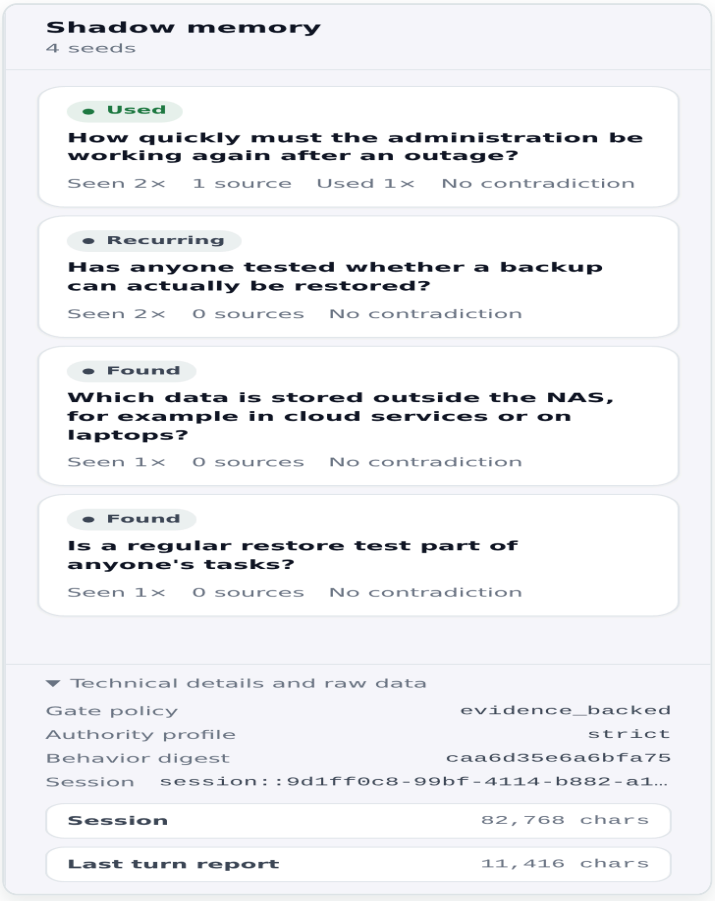


Figure 11 — Technical details of the example conversation.

10 Contradicting a seed

Is a seed wrong, or out of date? Open it and click **Mark as contradicted** under **Contradiction**. The seed becomes **Blocked** and can no longer influence any answer, even if it was promoted.

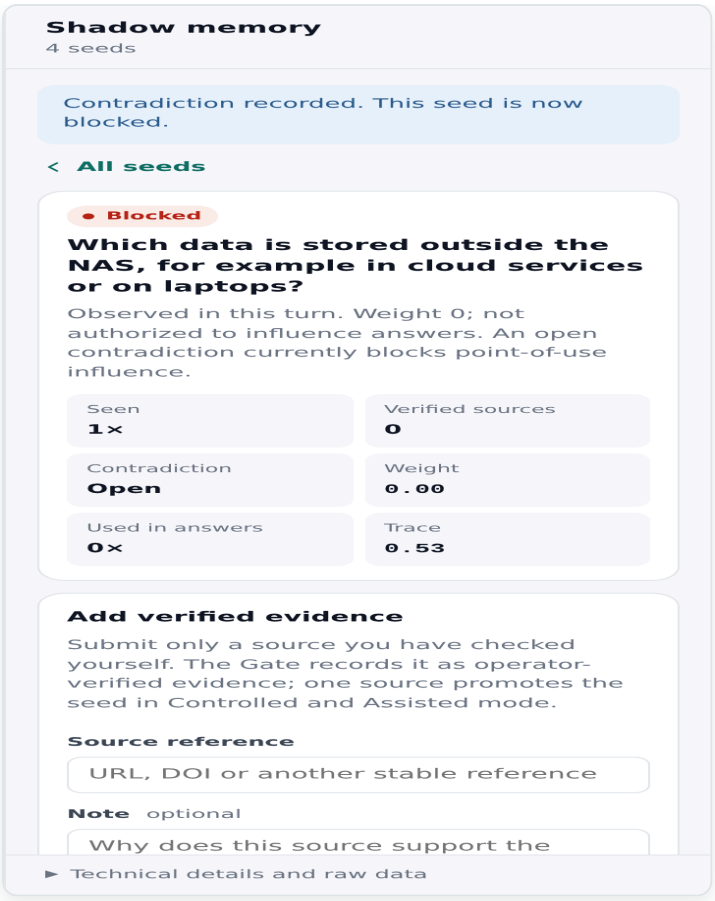


Figure 12 — A blocked seed.

To lift the block, enter under **Basis for resolution** why the contradiction no longer applies, and click **Resolve contradiction**. That removes only the block. The seed does not get its weight back by itself: that needs a new verified source.

11 Authority modes

The authority mode decides how a seed can gain influence. You choose it when you start a conversation; after that it is fixed.

Mode	How a seed gains influence	Use it for
Controlled (default)	Only through a source you checked. One source is enough.	Everyday work
Assisted	As Controlled. In addition, Shadowseed points out seeds that keep coming back without a source (label *Action required*).	When you want a nudge
Autonomous	Recurrence counts as support; no source is needed.	Research only

In Autonomous mode the system grants influence without you. Use it to study the mechanism, not for answers you want to rely on.

12 On a phone-sized screen

Shadowseed also works in a narrow window. The side panels then fold away: the menu at the top left opens your conversations, the **Shadow** button at the top right opens the Shadow memory.

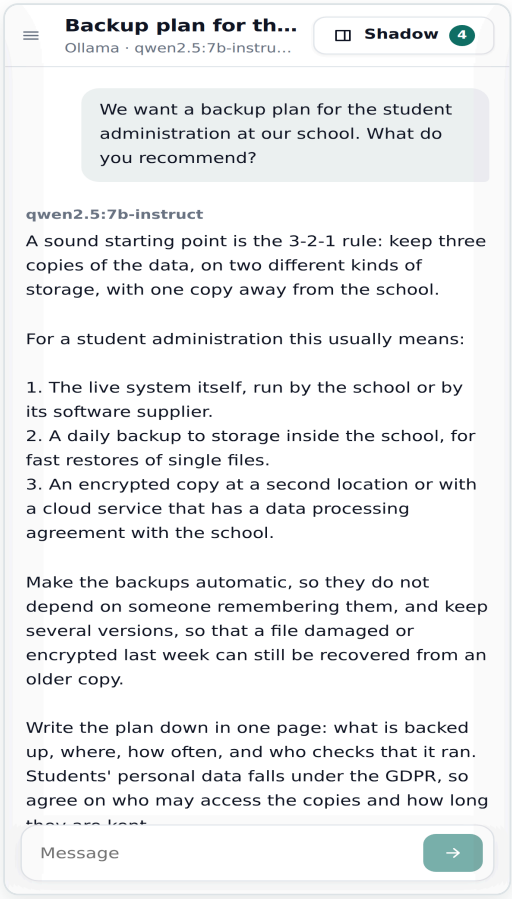


Figure 13 — The example conversation on a phone-sized screen.

Shadowseed always runs on the computer where you started it. A phone cannot reach it over the network, so this layout is for a narrow window on the same computer.

13 Using OpenAI

The downloadable app works with local models only. If you install Shadowseed as a Python package with the `openai` extra (chapter 21), you can also use OpenAI models:

1. Open **Providers** at the bottom left and enter a **Temporary API key**. The key stays in memory only and disappears when Shadowseed stops.
2. When you create a conversation, choose *OpenAI (hosted)* and confirm that your questions, seeds and texts are sent to OpenAI.
3. You confirm this again with every message.

> **Caution.** With OpenAI your data leaves your computer. What OpenAI does with it is governed by your own agreement with OpenAI.

14 Data, backups and privacy

What Shadowseed stores

Everything is in the `.shadowseed` folder in your home folder:

System	Folder
Windows	`C:\Users\<name>\.shadowseed`
macOS	`/Users/<name>/.shadowseed`
Linux	`/home/<name>/.shadowseed`

It holds your conversations, the answers, the seeds, the Gate decisions, your source references and notes, and the audit ledger. Next to it is `.shadowseed-integrity`, with the key that protects the ledger. Nothing leaves your computer unless you use OpenAI.

> **Caution.** Do not put passwords, API keys or other secrets in conversations, source references or notes. Use made-up or anonymised data when real personal data is not needed.

Making and restoring backups

The app has no backup button yet. You use the Python package from the same release (chapter 21):

Command	What it does
`shadowseed workspace info`	Shows where your data is and how much there is
`shadowseed workspace backup --output <file>`	Makes a complete, consistent copy
`shadowseed workspace restore <file>`	Restores a copy
`shadowseed workspace delete --yes`	Deletes everything in `.shadowseed` and `.shadowseed-integrity`

> **Caution.** Always give `backup` a file outside `.shadowseed`. Without `--output` the backup goes to `.shadowseed/exports`, and that folder is deleted along with the workspace.

Do not copy or move the folders by hand. A backup holds the same data as the original; keep it just as carefully.

Exporting a conversation

Command	Contents
`shadowseed export-workbench-report <session>`	The whole conversation, with checksums. Contains all text
`shadowseed export-support-bundle <session>`	Structure and counts only, without conversation or seed text. Meant for problem reports

The session id is under **Technical details and raw data**, next to **Session**.

15 Troubleshooting

What you see	Likely cause	Solution
The browser does not open	No default browser, or blocked	Copy the address from the Shadowseed window
macOS: "cannot be opened"	Quarantine on a downloaded app	Right-click → Open, or the `xattr` command from chapter 4
Windows: SmartScreen warning	App not signed by Microsoft	*More info* → *Run anyway*
"Ollama not reachable"	Ollama is not running	Start Ollama and try again
Message about `embeddinggemma`	Model missing or Ollama too old	`ollama pull embeddinggemma`; update Ollama
"Model not found"	Name differs from the installed name	Copy the name exactly from `ollama list`
An answer takes minutes	The model runs on the processor	Normal without a GPU; choose a smaller model
No seeds appear	Small models find little	Continue the conversation, or choose a larger model
Label *Provider needed*	This conversation's model is unreachable	Start Ollama, or set up OpenAI

Shadowseed doctor

The Python package includes a diagnostic program. It checks the workspace, the database, the audit ledger, permissions, disk space and Ollama, and tells you how to fix what it finds:

```
$ shadowseed doctor
python                  OK    3.11.15
workspace               OK    /home/you/.shadowseed
production_integrity    OK    verified ledger head sequence 7
ollama                  WARN  not detected at http://127.0.0.1:11434
                              Start `ollama serve`, pull a model, and rerun `shadowseed
doctor`.
Workbench can start.
```

The doctor is not part of the downloadable app. Install the Python package from the same release (chapter 21); it checks the same `.shadowseed` folder.

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16 Limitations

- **Few seeds with small models.** With a 7B model a seed appears in about one turn in eight, and about a quarter of those are useful.
- **Slow without a GPU.** Two to three minutes per answer with a 7B model.
- **No evidence of better answers.** Whether SSL improves answers is still being studied.
- **Seeds can be wrong.** A promoted seed is supported by the source you chose, not proven.
- **One user, one computer.** Shadowseed is not a multi-user or network service.
- **Not signed by Apple or Microsoft.** Your system warns at first start.
- **English interface only** for now.
- **No compare button.** Setting an answer next to the same answer without SSL is possible only through the web API (chapter 19).
- **No backup button.** Backups are made with the Python package.

PART 4

Testing

For anyone who tries out a version and reports findings.

IN THIS PART

17 A test session

17 A test session

What to look at

- Can you download, start and use the app without other software?
- Is it always clear which model and which authority mode are active?
- Does it feel like an ordinary chat, even if you ignore the seeds?
- Do you understand the statuses and the explanation of a seed?
- Is it clear that you must check a source, and that recurrence is not evidence?
- Do the error messages help when Ollama or a model is missing?
- Can you close a conversation and reopen it later without losing anything?

How to test

1. Start with the offline demo. It shows the controls, not the quality.
2. Then switch to Ollama with a real model and hold conversations of eight turns or more. Seeds and recurrence need time.
3. For each session, note the model, the authority mode and what you did.

Reporting a problem

A good report contains:

- what you were trying to do;
- the model and the authority mode;
- the shortest series of steps that brings the problem back;
- what you expected and what happened;
- whether a seed was used in that turn.

Leave out personal data and secrets. Attach a support bundle rather than a full report (chapter 14).

> **Important.** If anyone can change a seed's weight or status without going through the Gate, that is a serious defect. Report it with priority.

Collecting support bundles

A support bundle holds structure and counts but no conversation or seed text. Whoever coordinates a test with several testers combines bundles with `scripts/aggregate_support_bundles.py`. Do not edit a bundle by hand: the integrity check then fails.

PART 5

Research

The model behind Shadowseed, and how to study it fairly.

IN THIS PART

18 The SSL model

19 Showing that SSL had an effect

20 Studies and evidence

18 The SSL model

Lifecycle

A seed passes through fixed steps. Each step is separate from the others, and only the Gate changes authority.

Step	What happens
Detection	The model reads the question, the draft answer and at most four earlier turns, and proposes zero to five open points (prompt `detector_current_pair` v0.6)
Intake	A proposal becomes a seed with weight 0 if it is one question, or one English statement that is not a bare list
Recurrence	The session model judges whether a new note is the same open question as an earlier one (ADR-012). Text similarity serves as a floor (0.3) and as the fallback
Trace	Records recent presence and decays with a half-life. Seeds become *Dormant* and later *Expired*
Gate	The only place where weight and status change
Point of use	The last check just before an answer

Gate policy

Policy	Accepts
`evidence_backed`	Verified external support only. In Controlled and Assisted one source promotes (threshold 0.2, ADR-011)
`exploratory`	Enough recurrence, or verified external support
`legacy_evidence_required`	Historical thresholds, kept for comparability

A source needs a non-empty `source_ref`. That reference is the identity of the evidence: the same reference never counts twice, not even through another route. A contradiction is a separate record with an id, reason, status and basis for resolution. Resolving lifts the block but restores no weight.

Point of use

Before a seed may influence an answer, `AgentSafetyContract.decide_and_record` checks four things: positive weight, status *promoted*, no blocking contradiction, and a valid Gate decision for the seed's current authority version. Both permission and refusal are recorded, and every permission refers to the Gate decision it rests on. That makes it possible to see afterwards whether an outdated permission was used.

Two kinds of memory

Under `evidence_backed`, recurrence is only a trace: an observation you can study. Under `exploratory`, recurrence can count as support after three occurrences. The product uses `evidence_backed` by default.

19 Showing that SSL had an effect

Two answers always differ a little, even without SSL. A different answer is therefore no proof that a seed had influence. Attribute an effect to SSL only when:

1. the seed was promoted;
2. it was selected as relevant for that turn;
3. the point-of-use check allowed it;
4. it was actually injected, as recorded in the turn report and the timeline.

For a fair comparison, send a turn through the web API (chapter 24) with `compare_without_ssl: true`. Shadowseed then produces a second answer with the same model, the same visible history and the same question, but without seeds. The comparison changes nothing in the stored state. You find it under **Technical details and raw data → Last turn comparison**.

20 Studies and evidence

The repository contains pre-registered studies and their results:

Study	Question	Status
Seed calibration (`docs/research/seed-calibration.md`)	How useful are the seeds found, and how reliable is recurrence recognition?	Completed
Detector calibration (`docs/research/detector-calibration.md`)	Does a different detector prompt improve Qwen 2.5 7B's seeds?	Completed: more seeds, not better ones
Evidence efficacy (`docs/research/evidence-efficacy.md`)	Does an authorised seed improve answers, compared with SSL off and with self-critique?	Pilot

Four principles apply to every study:

- record the design and the decision rule before measuring;
- review blind, and record the ratings before opening the key;
- record the exact model version, the source commit and the prompt versions;
- treat what the model says as observation, never as evidence.

Research tools are in the `research/` package and in `scripts/`. Results are stored with their provenance under `benchmarks/`.

PART 6

Development

Working with the source code, the structure of the system and its interfaces.

IN THIS PART

21 Working from source

22 Architecture

23 Command line

24 Web API

25 Builds and releases

21 Working from source

Installing

From a release, without Git:

```
python -m pip install shadowseed-0.12.4-py3-none-any.whl
```

From the repository:

```
git clone https://github.com/E-AI-MODEL/shadowseed-pro.git
cd shadowseed-pro
python -m pip install -e ".[test]"
```

Python 3.10 or later is required. Optional extras:

Extra	Adds
`openai`	OpenAI models and embeddings
`models`	Hugging Face and Sentence Transformers
`workbench`	The legacy Gradio interface
`vector`, `build`, `standalone-build`, `security`	Vector storage, builds and security checks
`dev`	All of the above

Tests and checks

```
python -m pytest -q            # full test suite
ruff check src tests scripts   # style and errors
uv lock --check                # lockfile matches pyproject.toml
```

The web client is a Next.js app in `apps/web`. Build it and copy the output into the Python package with `scripts/sync_web_assets.py`.

Rules for changes

- The Validation Gate is the only route to authority. Do not add a second one.
- The fields `weight`, `status`, `evidence_count` and `contradiction_score` cannot be assigned directly; tests guard this.
- A change to the authority model needs an ADR in `docs/architecture/adr/`.
- Every change goes through a pull request with green CI.

22 Architecture

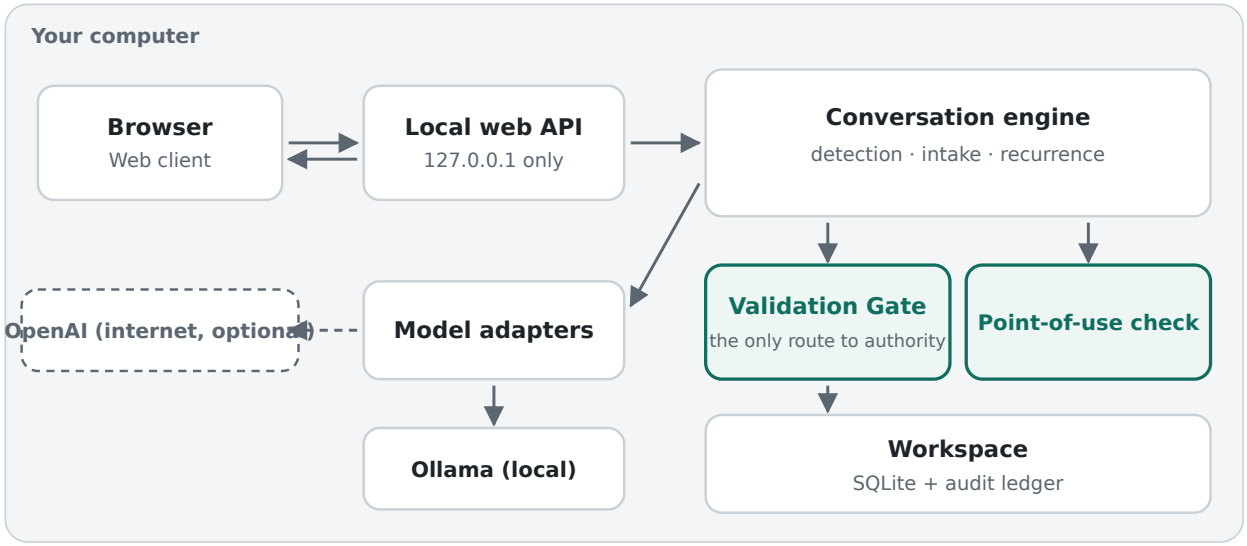


Figure 14 — The parts of Shadowseed. Everything runs on your own computer; OpenAI is optional.

Part	Location	Role
Conversation engine	`src/shadowseed/` (`chat.py`, `manager.py`, `lifecycle.py`)	Conversation, seeds and lifecycle
Detection	`src/shadowseed/detection/`	Detector prompts and parsing
Intake and recurrence	`intake.py`, `recurrence_*.py`	Which proposals become seeds, and when a seed recurs
Gate	`src/shadowseed/gate/`	The Validation Gate and its ledger
Storage	`src/shadowseed/storage/`	SQLite workspace and integrity ledger
Application	`src/shadowseed/application/`	Workspace, limits, health checks
Adapters	`src/shadowseed/adapters/`	Ollama, OpenAI, Hugging Face, embeddings
Web API	`src/shadowseed_webapi/`	Loopback HTTP server and launcher
Web client	`apps/web/`	The browser interface
Legacy Workbench	`src/shadowseed/workbench/`	Gradio interface, on request only
Research	`research/`	Benchmarks and studies

Design decisions are in `docs/architecture/adr/` (ADR-001 to ADR-012). Start with ADR-001 (the Gate as the sole source of authority), ADR-006 (the boundary of the local production profile), ADR-011 (one-source promotion) and ADR-012 (model-recognised recurrence).

23 Command line

Command	Purpose
`shadowseed-web [--port N] [--workspace DIR] [--no-browser]`	Start the app (default port 8765, loopback only)
`shadowseed-web --self-test --workspace DIR`	Run the built-in self-test
`shadowseed chat --backend fixture|ollama|openai`	Chat in the terminal
`shadowseed doctor [--json] [--workspace DIR]`	Check the installation
`shadowseed init [--workspace DIR]`	Create a workspace
`shadowseed workspace info|backup|restore|delete`	Manage the workspace
`shadowseed-workbench [--legacy-gradio]`	Start the desktop launcher; `--legacy-gradio` opens the old interface
`shadowseed export-workbench-report <session>`	Export a conversation as a full report
`shadowseed export-support-bundle <session>`	Export a support bundle without text

Every command shows all its options with `--help`. `docs/usage/cli.md` describes the research and benchmark commands.

24 Web API

The server listens on `127.0.0.1` only. All routes start with `/api/v1`.

Method	Route	Purpose
GET	`/health`	Health check
GET	`/providers`	Status of the model providers
POST	`/providers/openai/credential`	Set a temporary OpenAI key
POST	`/providers/openai/credential/clear`	Clear that key
GET	`/sessions`	List conversations
POST	`/sessions`	Create a conversation (`title` , `backend` , `model_id` , `authority_mode` , `external_confirmed`)
GET	`/sessions/{id}`	Read a conversation

Method	Route	Purpose
POST	`/sessions/{id}/turns`	Send a message (`question` , `request_id` , optional `compare_without_ssl` , `external_confirmed`)
GET	`/sessions/{id}/seeds/{seed}`	Read a seed with its timeline
POST	`/sessions/{id}/seeds/{seed}/evidence`	Record a source (`source_ref` , `note` , `operator_verified: true` , `request_id`)
POST	`/sessions/{id}/seeds/{seed}/contradictions`	Record a contradiction (`request_id`)
POST	`/sessions/{id}/seeds/{seed}/contradictions/resolve`	Resolve a contradiction (basis required, `request_id`)

Safe retries. A turn and the three seed actions take a `request_id` . Sending the same id again is safe: the action happens once. An id reused with different content is refused. Creating a conversation and setting or clearing the OpenAI key take no `request_id` ; repeating them repeats the action.

Isolation. Requests from another website, or with a `Host` header that does not point to loopback, are refused.

25 Builds and releases

Step	What happens
CI on every pull request	Tests on Python 3.10 and 3.12, local production checks on Windows, macOS and Linux, a strict audit of the shipped dependencies, and frozen builds for the three platforms
Release Workbench (manual)	Publishes the version in `pyproject.toml` from the exact commit on `main` , after CI on that commit is green. Builds the three downloads, `SHA256SUMS` , `PROVENANCE.json` , an SBOM, the wheel and the sdist
Production Release Assurance (automatic)	Checks the attestations, checksums and provenance of the published tag

A version is called *production-ready/local* only when a real use cycle with Ollama and `shadowseed doctor` on the published version has also been recorded, with no open serious findings (P0/P1). Release notes are in `docs/workbench/release-<version>.md` .

PART 7

Security and operations

What the local profile does and does not protect, and how to run Shadowseed reliably.

IN THIS PART

26 Security model

27 Running reliably

28 Settings

26 Security model

Shadowseed claims	Shadowseed does not claim
The server is reachable only through `127.0.0.1`, and the `Host` header is checked against DNS rebinding	Protection against a compromised operating system or administrator
Actions that change authority are attributed to a trusted local user	Separation between users, or safety on a hostile network
A client flag alone never grants authority	That a source the user marks as checked is correct
The Gate and the point-of-use check are the only route to influence	General protection against prompt injection
Tampering with the history, or rolling back to an older state, is detectable	Better answers in general
Operational logs contain as little content as possible	Secure deletion beyond what the file system provides
Releases are tied to an exact commit through checksums and provenance	

The full threat model is in `docs/architecture/production-threat-model.md`.

27 Running reliably

- **Limits.** Oversized input is refused before anything is stored. A model call stops after 120 seconds and is not retried automatically. A failed turn does not change the stored state.
- **Permissions.** On macOS and Linux only the owner can read the workspace. On Windows the user account protects it.
- **Integrity.** The audit ledger records every authority event, protected by the key in `.shadowseed-integrity`. Never repair a workspace by deleting that folder; restore a verified backup instead.
- **Backups.** Use `shadowseed workspace backup --output <file>` with a file outside the workspace (chapter 14).
- **Containers.** `Dockerfile.workbench` is meant for development. Publish the port on loopback only: `-p 127.0.0.1:7860:7860`.
- **Diagnosis.** `shadowseed doctor` tells a usable state apart from integrity problems and missing parts.

Operating procedures: `docs/operations/production-local-recovery.md` and `docs/operations/production-local-release.md`.

28 Settings

Setting	Default	Meaning
`--port`	8765	Preferred port; if it is taken, Shadowseed picks another
`--workspace`	`~/.shadowseed`	The data folder
`OLLAMA_HOST`	`http://localhost:11434`	Address of Ollama
`OPENAI_API_KEY`	not set	OpenAI key for the Python package; the app uses the temporary key
Authority mode	Controlled	Chosen per conversation
Embedding model	`embeddinggemma`	Needed for conversations with Ollama

Appendices

IN THIS PART

A Glossary

B Frequently asked questions

C About this document

A Glossary

Term	Meaning
Seed	A note about a possibly open point in the conversation
Shadow memory	All seeds of a conversation with their history
Validation Gate	The only place where a seed gains or loses influence
Authority mode	Controlled, Assisted or Autonomous
Authority	A seed's right to influence answers
Weight	How much authority the Gate granted, from 0 to 1
Trace	How recently a seed was present
Recurrence	An open point that comes back in a later turn
Verified evidence	A source the user has read
Contradiction	A recorded objection that blocks a seed
Point of use	The last check just before an answer
Gate policy	`evidence_backed` or `exploratory`
Workspace	The `.shadowseed` data folder
Audit ledger	The protected record of all authority events
Offline demo	Working without a model, to explore the controls

B Frequently asked questions

Does Shadowseed send my conversations anywhere? No, unless you choose OpenAI. With Ollama everything stays on your computer.

Why do I see so few seeds? Small models find few open points. Longer conversations and larger models help.

I added a source, but the seed is not used. Why? A promoted seed takes part only when it fits your question. Ask a question about the seed's topic.

Can I set a seed's weight myself? No. Only the Gate changes authority. That is a deliberate safeguard.

Is a seed that keeps coming back promoted automatically? Not in Controlled and Assisted. There a verified source is always needed. Only in Autonomous does recurrence count.

Is a promoted seed true? Not necessarily. It is supported by the source you checked.

May I use Shadowseed in my company? The licence is PolyForm Noncommercial 1.0.0. Commercial use needs separate permission.

C About this document

Applies to	Shadowseed 0.12.4
Editions	English and Dutch, as Markdown and as PDF in `docs/manual/`
Screenshots	Shadowseed 0.12.4 with a pre-written example conversation (see chapter 2)
Licence	PolyForm Noncommercial License 1.0.0

Found an error or have a suggestion? Report it as an issue in the repository.